Understanding the NS Blow-Up-Factor Proof

Machine working note

August 18, 2026

1 Free-field realization

This note follows the proof in arXiv:1312.4520, but uses standard physicists' notation. We denote the bosonic oscillators by a_n, the NS fermion by ψ_r, and the bosonic zero modes by $\widehat{p}, \widehat{q}$. Thus a_n is the oscillator denoted by c_n in the paper.

Set

$$Q = b + b^{-1}, \qquad c_{\mathrm{NS}} = \frac{3}{2} + 3Q^2. \tag{1}$$

1.1 Oscillators and Fock states

The free fields satisfy

$$[a_m, a_n] = m\delta_{m+n,0}, \qquad \{\psi_r, \psi_s\} = \delta_{r+s,0}, \qquad [\widehat{p}, \widehat{q}] = -i, \tag{2}$$

where $m, n \in \mathbb{Z} \setminus \{0\}$ and $r, s \in \mathbb{Z} + \frac{1}{2}$. The momentum-$p$ Fock vacuum obeys

$$\widehat{p}|p\rangle = p|p\rangle, \qquad a_n|p\rangle = 0, \qquad \psi_r|p\rangle = 0, \qquad n, r > 0. \tag{3}$$

The corresponding Fock space is denoted by $\mathcal{H}_p$.

1.2 The I-pairing and BPZ pairing

We distinguish the free-field Hermitian conjugation I from BPZ conjugation $\dagger$. The former is the antilinear anti-involution defined by

$$(\lambda X)^{\mathsf{I}} = \overline{\lambda}\, X^{\mathsf{I}}, \qquad (XY)^{\mathsf{I}} = Y^{\mathsf{I}} X^{\mathsf{I}}, \qquad \widehat{p}^{\mathsf{I}} = \widehat{p}, \qquad \widehat{q}^{\mathsf{I}} = \widehat{q}, \qquad a_n^{\mathsf{I}} = a_{-n}, \qquad \psi_r^{\mathsf{I}} = \psi_{-r}. \tag{4}$$

In the notation of the human note,

$$c_n = a_n, \qquad \eta_r = \psi_r, \qquad \widehat{P} = i\widehat{p}. \tag{5}$$

Therefore Eq. (4) is equivalently

$$c_n^{\mathsf{I}} = c_{-n}, \qquad \eta_r^{\mathsf{I}} = \eta_{-r}, \qquad \widehat{P}^{\mathsf{I}} = -\widehat{P}. \tag{6}$$

The auxiliary fermion denoted by ψ_r in the human note is denoted by f_r in this machine note. When it is introduced below, its convention is

$$f_r^{\mathsf{I}} = -f_{-r}, \qquad \text{equivalently} \qquad \psi_r^{\mathsf{I}} = -\psi_{-r} \quad \text{in the notation of the human note.} \tag{7}$$

We denote the corresponding free-field Hermitian product by $\langle \cdot, \cdot \rangle_{\mathsf{I}}$, and write

$$ {}_{\mathsf{I}}\langle u| := (|u\rangle)^{\mathsf{I}}. \tag{8} $$

By contrast, $\dagger$ is reserved for BPZ conjugation on an abstract NS module:

$$ L_n^\dagger = L_{-n}, \qquad G_r^\dagger = G_{-r}. \tag{9} $$

For real Q and p, the free-field conjugation is compatible with the two realizations below:

$$ \big(L_n^{(\epsilon)}\big)^{\mathsf{I}} = L_{-n}^{(\epsilon)}, \qquad \big(G_r^{(\epsilon)}\big)^{\mathsf{I}} = G_{-r}^{(\epsilon)}. \tag{10} $$

Thus both pairings give the same adjoint relations for the SCA generators, but they are defined in different places: $\dagger$ acts on the abstract NS module, whereas I acts on the free-field realization. If $\iota_\epsilon(p)$ is the module identification normalized by $\iota_\epsilon(p)|\Delta_p\rangle = |\epsilon p\rangle$, then on the real momentum slice it may be chosen so that

$$ \langle u|v\rangle_{\mathrm{BPZ}} = {}_{\mathsf{I}}\langle \iota_\epsilon(p)u|\iota_\epsilon(p)v\rangle. \tag{11} $$

The equality follows because the two sides have the same normalized highest-weight matrix element and obey the same recursion under L_n and G_r. Accordingly, the abstract matrix elements on the left-hand side of the comparison below use the BPZ pairing, while their free-field representatives on the right-hand side use the I-pairing. This compatibility does not identify I with BPZ conjugation, especially after analytic continuation in the momenta.

1.3 Super-Virasoro generators

On $\mathcal{H}_p$ there are two reflected free-field realizations, labelled by $\epsilon = \pm 1$:

$$ L_0^{(\epsilon)} = \sum_{m=1}^{\infty} a_{-m}a_m + \sum_{r\geq\frac{1}{2}} r\psi_{-r}\psi_r + \frac{Q^2}{8} + \frac{p^2}{2}, \tag{12} $$

$$ L_n^{(\epsilon)} = \frac{1}{2}\sum_{m\neq 0,n} a_{n-m}a_m + \frac{1}{2}\sum_{r\in\mathbb{Z}+\frac{1}{2}} r\psi_{n-r}\psi_r + \left(\frac{inQ}{2} + \epsilon p\right) a_n, \qquad n \neq 0, \tag{13} $$

$$ G_r^{(\epsilon)} = \sum_{m\neq 0} a_m\psi_{r-m} + (iQr + \epsilon p)\,\psi_r. \tag{14} $$

They satisfy the NS super-Virasoro algebra with central charge (1). In either realization, $|p\rangle$ is a highest-weight state of weight

$$ \Delta_p = \frac{Q^2}{8} + \frac{p^2}{2}. \tag{15} $$

For most of the proof one chooses one realization and writes it simply as $L_n(p), G_r(p)$. The second realization becomes important when the reflection map is introduced.

1.4 The chiral exponential vertex

Split the chiral boson into creation and annihilation parts,

$$ \varphi_<(z) = -i\sum_{n=1}^{\infty} \frac{a_{-n}}{n} z^n, \qquad \varphi_>(z) = i\sum_{n=1}^{\infty} \frac{a_n}{n} z^{-n}, \tag{16} $$

and define

$$\psi(z) = \sum_{r\in\mathbb{Z}+\frac{1}{2}} \psi_r z^{-r-\frac{1}{2}}. \tag{17}$$

The free-field vertex operator of momentum α is the ordered exponential

$$\boxed{E^{\alpha}(z) = z^{-\Delta_\alpha} e^{\frac{\alpha}{2}\widehat{q}} e^{\alpha\varphi_<(z)} z^{-i\alpha\widehat{p}} e^{\alpha\varphi_>(z)} e^{\frac{\alpha}{2}\widehat{q}}}, \qquad \Delta_\alpha = \frac{1}{2}\alpha(Q-\alpha). \tag{18}$$

In physicists' shorthand,

$$E^{\alpha}(z) = z^{-\Delta_\alpha} : e^{\alpha\varphi(z)} :, \tag{19}$$

where the normal-ordering symbol includes the symmetric zero-mode ordering displayed explicitly in Eq. (18).

The two basic checks are

$$[\widehat{p}, E^{\alpha}(z)] = -i\alpha E^{\alpha}(z), \tag{20}$$

$$[L_n(p), E^{\alpha}(z)] = z^n \left(z\partial_z + (n+1)\Delta_\alpha\right) E^{\alpha}(z), \tag{21}$$

$$[G_r(p), E^{\alpha}(z)] = -i\alpha z^{r+\frac{1}{2}} \psi(z) E^{\alpha}(z). \tag{22}$$

Hence $E^{\alpha}(z)$ is the free-field representative of an NS primary vertex and shifts the Fock momentum according to

$$E^{\alpha}(z) : \mathcal{H}_p \longrightarrow \mathcal{H}_{p-i\alpha}. \tag{23}$$

At this stage no screening charge has been introduced. The ordered exponential is the local chiral primary vertex; screening charges will enter only when we ask for a nonzero matrix element between Fock spaces whose momenta do not satisfy the neutrality condition for E^{α} alone.

2 Why the screened matrix element represents an NS primary

The goal of this section is to explain the identity written as Eq. (2.9) in arXiv:1312.4520. The logic has three steps:

1. define a screened matrix element in the free-field Fock space and determine its SCA properties;
2. define the corresponding matrix element of an abstract NS primary chiral vertex;
3. compare the two constructions and fix the normalization for which they agree exactly.

2.1 Screening charges

For either

$$\beta = b \qquad \text{or} \qquad \beta = b^{-1}, \tag{24}$$

one has

$$\Delta_\beta = \frac{1}{2}\beta(Q-\beta) = \frac{1}{2}. \tag{25}$$

Consequently,

$$S_\beta(z) := \psi(z) E^{\beta}(z) \tag{26}$$

is a fermion-odd field of conformal weight one. Its transformations are total derivatives:

$$[L_n(p), S_\beta(z)] = \partial_z\Big(z^{n+1}S_\beta(z)\Big)\,, \tag{27}$$

$$\{G_r(p), S_\beta(z)\} = \frac{i}{\beta}\partial_z\Big(z^{r+\frac{1}{2}}E^\beta(z)\Big)\,. \tag{28}$$

For a closed Coulomb-gas contour, define

$$\mathcal{Q}_\beta := \oint dz\, S_\beta(z) = \oint dz\, \psi(z)E^\beta(z). \tag{29}$$

The total derivatives imply

$$[L_n(p), \mathcal{Q}_\beta] = 0, \qquad \{G_r(p), \mathcal{Q}_\beta\} = 0. \tag{30}$$

Thus a screening charge shifts the Fock momentum while supercommuting with the SCA:

$$\mathcal{Q}_b : \mathcal{H}_p \longrightarrow \mathcal{H}_{p-ib}, \qquad \mathcal{Q}_{1/b} : \mathcal{H}_p \longrightarrow \mathcal{H}_{p-i/b}. \tag{31}$$

2.2 Step 1: the screened free-field matrix element

Fix nonnegative integers r, s. For the even NS primary component, assume

$$r + s \in 2\mathbb{Z}, \tag{32}$$

so that the product of screening charges is even. Define the screened free-field operator

$$\mathcal{I}_{\alpha;r,s}(z) := E^\alpha(z)\mathcal{Q}_b^r\mathcal{Q}_{1/b}^s \tag{33}$$

and its matrix element

$$\mathcal{F}^{\pm,\pm}_{\alpha;r,s}(q, p; z) := \langle \pm q|E^\alpha(z)\mathcal{Q}_b^r\mathcal{Q}_{1/b}^s| \pm p\rangle. \tag{34}$$

The signs in front of p and q are independent.

The operator in Eq. (33) shifts the momentum by

$$-i\left(\alpha + rb + sb^{-1}\right). \tag{35}$$

Therefore Eq. (34) can be nonzero only on the neutrality locus

$$\boxed{\pm q = \pm p - i\alpha - i\left(rb + sb^{-1}\right)}. \tag{36}$$

The screening charges do not alter the SCA transformation law of E^α. Hence

$$[L_n(p), \mathcal{I}_{\alpha;r,s}(z)] = z^n\left(z\partial_z + (n+1)\Delta_\alpha\right)\mathcal{I}_{\alpha;r,s}(z), \tag{37}$$

where

$$\Delta_\alpha = \frac{1}{2}\alpha(Q - \alpha). \tag{38}$$

Define the odd partner

$$\mathcal{I}^*_{\alpha;r,s}(z) := -i\alpha\,\psi(z)E^\alpha(z)\mathcal{Q}_b^r\mathcal{Q}_{1/b}^s. \tag{39}$$

Then

$$[G_r(p), \mathcal{I}_{\alpha;r,s}(z)] = z^{r+\frac{1}{2}} \mathcal{I}^*_{\alpha;r,s}(z), \tag{40}$$

$$[L_n(p), \mathcal{I}^*_{\alpha;r,s}(z)] = z^n \left(z\partial_z + (n+1) \left(\Delta_\alpha + \frac{1}{2} \right) \right) \mathcal{I}^*_{\alpha;r,s}(z), \tag{41}$$

$$\{G_r(p), \mathcal{I}^*_{\alpha;r,s}(z)\} = z^{r-\frac{1}{2}} \left(z\partial_z + (2r+1)\Delta_\alpha \right) \mathcal{I}_{\alpha;r,s}(z). \tag{42}$$

These are precisely the transformation laws of an NS primary supermultiplet.

In particular, the L_0 Ward identity fixes the coordinate dependence:

$$\mathcal{F}^{\pm,\pm}_{\alpha;r,s}(q,p;z) = z^{\Delta_q - \Delta_\alpha - \Delta_p} \mathcal{F}^{\pm,\pm}_{\alpha;r,s}(q,p;1), \tag{43}$$

where

$$\Delta_p = \frac{Q^2}{8} + \frac{p^2}{2}, \qquad \Delta_q = \frac{Q^2}{8} + \frac{q^2}{2}. \tag{44}$$

2.3 Step 2: the abstract NS-primary matrix element

Let $\mathcal{V}_{\Delta_p}$ and $\mathcal{V}_{\Delta_q}$ be NS super-Virasoro highest-weight modules. Their highest-weight states satisfy

$$L_0|\Delta_p\rangle = \Delta_p|\Delta_p\rangle, \qquad L_n|\Delta_p\rangle = G_r|\Delta_p\rangle = 0, \qquad n, r > 0, \tag{45}$$

$$\langle\Delta_q|L_0 = \Delta_q\langle\Delta_q|, \qquad \langle\Delta_q|L_{-n} = \langle\Delta_q|G_{-r} = 0, \qquad n, r > 0. \tag{46}$$

We use unit normalization in each fixed module:

$$\langle\Delta_p|\Delta_p\rangle = \langle\Delta_q|\Delta_q\rangle = \langle\Delta_\alpha|\Delta_\alpha\rangle = 1. \tag{47}$$

An even NS primary chiral vertex $V_{\Delta_\alpha}(z)$, together with its odd superpartner $V^*_{\Delta_\alpha}(z)$, is defined by

$$[L_n, V_{\Delta_\alpha}(z)] = z^n \left(z\partial_z + (n+1)\Delta_\alpha \right) V_{\Delta_\alpha}(z), \tag{48}$$

$$[G_r, V_{\Delta_\alpha}(z)] = z^{r+\frac{1}{2}} V^*_{\Delta_\alpha}(z), \tag{49}$$

$$[L_n, V^*_{\Delta_\alpha}(z)] = z^n \left(z\partial_z + (n+1) \left(\Delta_\alpha + \frac{1}{2} \right) \right) V^*_{\Delta_\alpha}(z), \tag{50}$$

$$\{G_r, V^*_{\Delta_\alpha}(z)\} = z^{r-\frac{1}{2}} \left(z\partial_z + (2r+1)\Delta_\alpha \right) V_{\Delta_\alpha}(z). \tag{51}$$

The primary matrix element is

$$\mathcal{A}_\alpha(q,p;z) := \langle\Delta_q|V_{\Delta_\alpha}(z)|\Delta_p\rangle. \tag{52}$$

Its L_0 Ward identity gives

$$\mathcal{A}_\alpha(q,p;z) = z^{\Delta_q - \Delta_\alpha - \Delta_p} \mathcal{A}_\alpha(q,p;1). \tag{53}$$

The remaining SCA Ward identities determine every descendant matrix element in terms of the single constant

$$C_{q\alpha p} := \langle\Delta_q|V_{\Delta_\alpha}(1)|\Delta_p\rangle. \tag{54}$$

The unit norms of the three primary states do not determine $C_{q\alpha p}$; it is the remaining normalization of the chiral vertex.

2.4 Step 3: comparison and exact normalization

Normalize the Fock vacua in each fixed-momentum fiber by

$$ {}_{|}\langle \pm p | \pm p \rangle = {}_{|}\langle \pm q | \pm q \rangle = 1. \tag{55}$$

Before restricting to fixed fibers, this is the usual delta-function normalization

$$ {}_{|}\langle p_1 | p_2 \rangle = \delta(p_1 - p_2). \tag{56}$$

Fix the module identifications by

$$\iota_\pm(p)|\Delta_p\rangle = |\pm p\rangle, \qquad \iota_\pm(q)|\Delta_q\rangle = |\pm q\rangle. \tag{57}$$

For generic real momenta these extend to unitary isomorphisms between the abstract NS Verma modules and their free-field realizations.

The screened operator $\mathcal{I}_{\alpha;r,s}$ and the abstract chiral vertex V_{Δ_α} now have

1. the same NS superconformal transformation laws,
2. the same conformal weight Δ_α,
3. the same fermion parity, and
4. compatible incoming and outgoing modules on the neutrality locus (36).

They therefore define the same chiral intertwiner up to one overall constant. Exact equality is obtained by choosing

$$\boxed{C_{q\alpha p} = \mathcal{F}^{\pm,\pm}_{\alpha;r,s}(q,p;1) = {}_{|}\langle \pm q | E^\alpha(1) \mathcal{Q}_b^r \mathcal{Q}_{1/b}^s | \pm p\rangle}. \tag{58}$$

With this normalization,

$$\boxed{\langle \Delta_q | V_{\Delta_\alpha}(z) | \Delta_p \rangle = {}_{|}\langle \pm q | E^\alpha(z) \mathcal{Q}_b^r \mathcal{Q}_{1/b}^s | \pm p\rangle}, \tag{59}$$

which is the first line of Eq. (2.9).

For descendant states $u \in \mathcal{V}_{\Delta_q}$ and $v \in \mathcal{V}_{\Delta_p}$, the normalization-independent form of the same statement is

$$\boxed{\frac{\langle u | V_{\Delta_\alpha}(1) | v\rangle}{\langle \Delta_q | V_{\Delta_\alpha}(1) | \Delta_p\rangle} = \frac{{}_{|}\langle \iota_\pm(q) u | E^\alpha(1) \mathcal{Q}_b^r \mathcal{Q}_{1/b}^s | \iota_\pm(p) v\rangle}{{}_{|}\langle \pm q | E^\alpha(1) \mathcal{Q}_b^r \mathcal{Q}_{1/b}^s | \pm p\rangle}}. \tag{60}$$

Thus the unit normalization of the states is necessary, but it is not sufficient by itself: one must also fix the remaining chiral-vertex constant as in Eq. (58).

2.5 The reflected exponential and the eight representations

Since

$$\Delta_{Q-\alpha} = \frac{1}{2}(Q-\alpha)\alpha = \Delta_\alpha, \tag{61}$$

the exponential $E^{Q-\alpha}$ has the same SCA primary weight. Its total momentum shift in the presence of the same screening charges is

$$-i\left(Q-\alpha+rb+sb^{-1}\right)=i\alpha-i\left((r+1)b+(s+1)b^{-1}\right). \tag{62}$$

The second neutrality locus is therefore

$$\boxed{\pm q=\pm p+i\alpha-i\left((r+1)b+(s+1)b^{-1}\right)}. \tag{63}$$

With the analogous choice of chiral-vertex normalization,

$$\boxed{\langle\Delta_q|V_{\Delta_\alpha}(z)|\Delta_p\rangle=\langle\pm q|E^{Q-\alpha}(z)\mathcal{Q}_b^r\mathcal{Q}_{1/b}^s|\pm p\rangle}. \tag{64}$$

The abstract weights depend only on p^2 and q^2, so the signs in front of p and q can be chosen independently. There are four sign choices and two exponential choices,

$$E^{\alpha}\qquad\text{or}\qquad E^{Q-\alpha}. \tag{65}$$

This gives the eight free-field representations stated below Eq. (2.9).

2.6 The precise meaning of Eq. (2.9)

Equation (2.9) is a statement about matrix elements on the charge-neutral loci (36) and (63). It is not one global operator identity

$$V_{\Delta_\alpha}(z)=E^{\alpha}(z)\mathcal{Q}_b^r\mathcal{Q}_{1/b}^s \tag{66}$$

valid for fixed r,s on every Fock module. Rather, each admissible screened expression realizes the same normalized chiral intertwiner on its own neutrality locus. Later, the blow-up-factor proof evaluates matrix elements on many such loci and uses polynomiality to continue the result to generic momenta.

For the even primary matrix element in Eq. (2.9), the total number $r+s$ of screening charges is even. For the odd superpartner matrix elements in Eqs. (2.10) and (2.11), it is odd.

3 From the screened intertwiner to Eq. (3.13)

We now explain the precise chain of substitutions that turns the normalized three-point block into the free-field matrix element in Eq. (3.13). The essential input is not an explicit formula for the reflected fermion. Instead, one first works in a free-field chart in which no reflected fermions occur.

3.1 The unreflected χ-states

Introduce the additional fermion f_r by

$$\{f_r,f_s\}=\delta_{r+s,0},\qquad f_r^{\dagger}=-f_{-r}, \tag{67}$$

and define

$$\chi_r=f_r-i\psi_r. \tag{68}$$

The two fermions anticommute with one another. It follows that

$$\chi_r^{\dagger} = -\chi_{-r}, \qquad \{\chi_r, \chi_s\} = 0. \tag{69}$$

Let

$$|p, 0\rangle := |p\rangle \otimes |0_f\rangle. \tag{70}$$

For $j > 0$, the double-Virasoro highest-weight state has the unreflected free-field representative

$$|p, j\rangle = \chi_{-j+\frac{1}{2}} \cdots \chi_{-\frac{3}{2}} \chi_{-\frac{1}{2}} |p, 0\rangle. \tag{71}$$

The pole-cleared state used in the three-point block is

$$|\xi_{p,j}\rangle := \Omega(p, j)|p, j\rangle, \tag{72}$$

where $\Omega(p, j)$ is the polynomial introduced in Eq. (2.24) of the paper. For real p, Q, b,

$$\Omega(p, j)^{\dagger} = \overline{\Omega(p, j)} = \Omega(-p, j). \tag{73}$$

Consequently, the free-field Hermitian bra associated with Eq. (72) is

$$(|\xi_{p,j}\rangle)^{\dagger} = \Omega(-p, j)\, {}_{\dagger}\langle p, 0| \chi^{\dagger}_{-\frac{1}{2}} \chi^{\dagger}_{-\frac{3}{2}} \cdots \chi^{\dagger}_{-j+\frac{1}{2}}. \tag{74}$$

The replacement $p \mapsto -p$ in the coefficient in Eq. (74) is caused by complex conjugation under $\dagger$. It is not an application of the reflection map.

3.2 Turning the middle state into contour insertions

The local field

$$\chi(\xi) = \sum_{r \in \mathbb{Z}+\frac{1}{2}} \chi_r \, \xi^{-r-\frac{1}{2}} \tag{75}$$

implements the χ-modes through the state-operator correspondence. Around an insertion at $z = 1$,

$$\chi_{-\ell+\frac{1}{2}} \mathcal{O}(1) = \frac{1}{2\pi i} \oint_1 \frac{d\xi}{(\xi - 1)^{\ell}} \chi(\xi) \mathcal{O}(1), \qquad \ell = 1, 2, \dots. \tag{76}$$

Therefore the field associated with the middle state can be written as

$$\begin{aligned} V_{\xi_{p_2,j_2}}(1) = \frac{\Omega(p_2, j_2)}{(2\pi i)^{j_2}} \oint_1 \frac{d\xi_{j_2}}{(\xi_{j_2} - 1)^{j_2}} \cdots \oint_1 \frac{d\xi_1}{\xi_1 - 1} \\ \times \chi(\xi_{j_2}) \cdots \chi(\xi_1) V_{p_2,0}(1). \end{aligned} \tag{77}$$

The contours and the fermionic fields are ordered as displayed. This ordering is inherited from the ordered product of modes in the state and must be kept fixed when signs are evaluated.

3.3 Replacing the primary vertex by a screened intertwiner

Choose a screening realization satisfying

$$p_3 = p_1 - i\left(\alpha_2 + rb + sb^{-1}\right), \tag{78}$$

and abbreviate

$$\mathcal{S}_{\alpha_2;r,s} := E^{\alpha_2}(1)\mathcal{Q}_b^r \mathcal{Q}_{1/b}^s, \qquad \mathcal{N}_{31} := {}_{\mathsf{I}}\langle p_3|\mathcal{S}_{\alpha_2;r,s}|p_1\rangle. \tag{79}$$

The Ward-identity argument of the previous section identifies the normalized abstract primary vertex with the normalized screened intertwiner:

$$\frac{V_{p_2,0}(1)}{\langle p_3|V_{p_2,0}(1)|p_1\rangle} \quad \longleftrightarrow \quad \frac{\mathcal{S}_{\alpha_2;r,s}}{\mathcal{N}_{31}}. \tag{80}$$

This is an equality of normalized chiral intertwiners on the neutrality locus, not a global operator identity. Once the primary matrix element is fixed, the SCA Ward identities guarantee equality of all descendant matrix elements.

3.4 Substitution into the normalized three-point block

Suppose $j_1, j_2, j_3 > 0$ and $j_1 + j_2 + j_3 \in 2\mathbb{Z}$. The normalized block is

$$\rho_{\mathrm{NS}}^{\mathcal{A}}\left(\xi_{p_3,j_3}, \xi_{p_2,j_2}, \xi_{p_1,j_1} \mid 1\right) := \frac{\langle \xi_{p_3,j_3}|V_{\xi_{p_2,j_2}}(1)|\xi_{p_1,j_1}\rangle}{\langle p_3|V_{p_2,0}(1)|p_1\rangle}. \tag{81}$$

The bras in Eq. (81) are abstract BPZ bras. After applying the normalized module identification, the right-hand free-field calculation uses the I-pairing.

Substituting Eqs. (71), (74), (77), and (80) gives

$$\begin{aligned}
&\rho_{\mathrm{NS}}^{\mathcal{A}}\left(\xi_{p_3,j_3}, \xi_{p_2,j_2}, \xi_{p_1,j_1} \mid 1\right)\\
&= \frac{\Omega(-p_3,j_3)\Omega(p_2,j_2)\Omega(p_1,j_1)}{(2\pi i)^{j_2}\mathcal{N}_{31}} \oint_1 \frac{d\xi_{j_2}}{(\xi_{j_2}-1)^{j_2}} \cdots \oint_1 \frac{d\xi_1}{\xi_1 - 1}\\
&\quad \times {}_{\mathsf{I}}\langle p_3, 0|\chi^{\mathsf{I}}_{-\frac{1}{2}} \cdots \chi^{\mathsf{I}}_{-j_3+\frac{1}{2}}\chi(\xi_{j_2}) \cdots \chi(\xi_1)\mathcal{S}_{\alpha_2;r,s}\\
&\qquad\qquad \times \chi_{-j_1+\frac{1}{2}} \cdots \chi_{-\frac{1}{2}}|p_1, 0\rangle.
\end{aligned} \tag{82}$$

This is Eq. (3.13), with the paper's free-field dagger replaced by our I. Expanding $\chi_r^{\mathsf{I}} = -\chi_{-r}$ produces the factor $(-1)^{j_3}$ that appears when the correlator is separated into its bosonic and fermionic parts.

3.5 What happened to the reflected fermion?

Equation (82) does not express $\psi_r^R(p)$ in terms of the original fermions. It avoids that problem:

1. For $j_1, j_2, j_3 > 0$, all three states are represented in the unreflected χ-basis.
2. Other sign choices are analyzed using the alternative free-field realizations associated with E^{α}, $E^{Q-\alpha}$, and the independent signs of the incoming and outgoing momenta.

3. The factors $\Omega(p_i, j_i)$ remove the possible poles and turn the normalized block into a polynomial of bounded degree.

4. Free-field calculations on the admissible screening loci determine the zeros and normalization of this polynomial, after which the result is continued to generic momenta.

Thus the authors circumvent the absence of a closed formula for the reflected fermion; they do not solve for it in the original oscillator basis. The calculation succeeds because each required specialization can be evaluated in a free-field chart adapted to that specialization.